

# Daily Scheduling of the Golden Spiny Mouse Under Photoperiodic and Social Cues

NAVA ZISAPEL,<sup>1\*</sup> EILON BARNEA,<sup>2</sup> IDO IZHAKI,<sup>2</sup> YOSSEI ANIS,<sup>1</sup> AND ABRAHAM HAIM<sup>2</sup>

<sup>1</sup>Department of Neurobiochemistry, Tel Aviv University, Tel Aviv, Israel 69978

<sup>2</sup>Department of Biology, University of Haifa at Oranim, Kiryat Tivon, Israel 36006

**ABSTRACT** A most important function of the circadian system is to ensure that behaviors and metabolism are appropriately timed with respect to the light/dark cycle and photoperiod. Ecological constraints can perturb the daily schedules; would they also impair photoperiodic adaptations? A natural model exists in the golden spiny mouse (*Acomys russatus*), which is nocturnal, but driven into diurnal activity when sharing the habitat with its congener, *A. cahirinus*. We show here that the presence of *A. cahirinus* alters the diurnal rhythms of body temperature and urine volume, delays excretion of the major melatonin metabolite, 6-sulfatoxymelatonin (6-SMT), and increases 2-deoxyglucose uptake by the suprachiasmatic nuclei in *A. russatus*. Nevertheless, a clear photoperiod effect on urine volume and 6-SMT rhythms was observed. These results indicate that the circadian system can adapt to major changes in daily scheduling without impairing daylength measurement, and consequently seasonal adaptation. *J. Exp. Zool.* 284:100–106, 1999. © 1999 Wiley-Liss, Inc.

The circadian system in mammals is characterized by a highly specific temporal order of various physiological and behavioral rhythms and appears to be similarly organized in nocturnal and diurnal species (Moore-Ede and Sulzman, '81). Thus, the activity rhythm of a nocturnal species such as the rat is approximately 180° out of phase with that of a diurnally active species such as the squirrel monkey (Sulzman et al., '78; Moore-Ede and Sulzman, '81). Yet, in both species the acrophases of temperature and urine production coincide with that of activity. An exception is melatonin production by the pineal gland, which is nocturnal regardless of activity mode. Melatonin serves as a daily signal of darkness and as a photoperiodic (daylength) cue. As such, this signal is used for the physiological and metabolic adaptation to the forthcoming season (Reiter, '91).

Ecological constraints can sometimes override the natural tendency to sleep at night or day and therefore dissociate the activity rhythms from the external day/night and the endogenous melatonin rhythm. Such a situation can be created artificially (e.g., shift work) and occurs naturally in the case of *Acomys russatus*. The golden spiny mouse, *A. russatus*, and the common spiny mouse, *A. cahirinus*, are two rodent species which coexist in the extremely hot environments of the south-

ern and the central parts of the Rift Valley in Israel (Harrison and Bates, '91). In the field, *A. cahirinus* is nocturnal whereas *A. russatus* is diurnal despite a greater risk of dehydration and hyperthermia. Removal of *A. cahirinus* from the common habitat resulted in a shift of *A. russatus* into nocturnal activity (Shkolnik, '71), thus indicating that interspecific relations determined the diurnal activity of the latter. Moreover, we have recently observed that chemical signals (odor of urine and feces) of *A. cahirinus* suffice to cause *A. russatus* to increase diurnal activity under laboratory conditions (Haim and Rozenfeld, '93).

How does the exposure to the competitor affect the daily metabolic adjustments and the ability of the animal to respond to change in daylength? To address these questions we have investigated the impact of short and long photoperiods on circadian rhythms of *A. russatus* in the absence and presence of its competitor *A. cahirinus*. In addition, we have assessed the major melatonin me-

Grant sponsor: USA-Israel Binational Science Foundation; Grant number: 91-00248.

\*Correspondence to: Nava Zisapel, Ph.D., Department of Neurobiochemistry, The George S. Wise Faculty of Life Sciences, Tel Aviv University, Tel Aviv 69978, Israel. E-mail: navazis@ccsg.tau.ac.il

Received 15 April 1998; Accepted 3 October 1998.

tabolite 6-sulfatoxymelatonin (6-SMT) in the urine as a marker for the endogenous pacemaker output. Glucose utilization, which provides an effective assay for oscillatory activity of the circadian pacemaker located in the suprachiasmatic nuclei (SCN) of the hypothalamus (Schwartz and Gainer, '77; Flood and Gibbs, '82), was also assessed.

We show here that in the presence of *A. cahirinus*, circadian rhythms in body temperature and urine volume were significantly modified whereas those of urinary 6-SMT and 2-DG uptake in the SCN were only slightly affected. These changes did not impair the ability of the animal to respond to changes in photoperiod.

## MATERIALS AND METHODS

### Animals

Nonsibling *A. russatus* males (mean body mass  $57 \pm 9$  g) were obtained from breeding colony (University of Haifa at Oranim) established by animals brought from the Dead Sea area and maintained 3–4 per cage. They were acclimated to long days, LD (16L:8D), or short days, SD (8L:16D), at 28°C for 3 weeks (light = cool white fluorescent illumination plus dim red light; dark = dim red light). Unless otherwise stated, lights were on between 06:00 and 22:00 hr in LD and 10:00 and 18:00 hr in SD.

### Body temperature measurements

Animals' body temperature ( $T_b$ ) was measured every 4 hr for 48 hr by use of a copper-constantan thermocouple connected to a TH-65 Wescor digital thermometer ( $\pm 0.01^\circ\text{C}$ ). The thermocouple was inserted 3 cm into the rectum for 30 sec. The mice were then exposed to *A. cahirinus* (1 animal/3 *A. russatus*) by placing the cages covered with a net with animals of both species close to each other for 5 days after which  $T_b$  was recorded every 4 hr for 30 hr. Although this procedure may be stressful to the animals, the impact of photoperiod and proximity to *A. cahirinus* on  $T_b$  rhythms of *A. russatus* measured by rectal thermometry resembles those measured by transmitters implanted into the abdominal cavity (Friedman et al., '97). Rectal measurements of  $T_b$  were thus preferred to avoid surgical procedures in these animals.

### Urine excretion volume measurements

Animals, treated as described above, were maintained in separate metabolic cages made of mesh nets and mounted above a parafilm sheet. Urine was collected every 4 hr for 48 hr before and af-

ter 5 days of exposure to *A. cahirinus*. Urine excretion volume (V) was calculated from the total volume excreted per collection, divided by the collection time interval.

### Urinary 6-sulfatoxymelatonin (6-SMT) assessment

Animals were treated as described above. Urine was collected separately from each animal every 4 hr for 48 hr before and after 5 days of exposure to *A. cahirinus*. Aliquots were assayed for 6-SMT (Stockgrand Ltd, Surrey, UK). The excretion rate of 6-SMT was calculated from the amount of metabolite excreted per collection, divided by the collection time interval.

### 2-Deoxyglucose uptake

Male *A. russatus* ( $n = 32$ ) were acclimated to LD or SD for three weeks at 28°C and then injected with  $^{14}\text{C}$ -2DG (200  $\mu\text{Ci/kg}$  in 50  $\mu\text{l}$  saline, i.p.). After 60 min these 16 mice were killed by decapitation, brains removed rapidly and frozen in dry ice. The other 16 animals acclimated to LD or SD were exposed to *A. cahirinus* (3 mice/16 *A. russatus* at adjacent cages in the same room) for 5 days and then injected with 2DG at the times specified above (4 mice per time point). The frozen brains were sliced by microtome. The slices containing the SCN were subjected to autoradiography (Kodak Direct Exposure) and autoradiographs analyzed by image analyzer. Extent incorporation into specific areas was transformed into artificial colors where red > yellow > green > blue. The location of the SCN in the sections used to generate the autoradiographs was verified by cresyl violet staining.

2DG injections of 4 animals at a time were performed at 10:00 or 18:00 hr in LD and at 11:00 or 14:00 hr in SD. We expected that, as in other rodents, glucose accumulation will be evident the SCN during subjective daytime only. At the times selected, the changes in  $T_b$ , urine excretion, and melatonin rhythms induced by exposure to *A. cahirinus* were found to be most pronounced. Thus, if a shift had occurred, either the "dusk" or "dawn" were supposed to become part of the subjective nighttime and thus SCN glucose uptake would be expected to decrease.

### Data evaluation

Periodic functions were fitted for each set of data (mean values of all individual animals) collected over a period of 24 or 48 hr. Whenever applicable, a cosinor function was fitted:

TABLE 1. Parameters of best fit regression of the baseline cosinor function<sup>1</sup>

| Parameter      | LD/SD | AC | M           | A24         | $\phi 24$  | A12         | $\phi 12$  |
|----------------|-------|----|-------------|-------------|------------|-------------|------------|
| T <sub>b</sub> | LD    | –  | 36.5 ± 0.13 | 0.83 ± 0.08 | 0.3 ± 0.5  | 0           |            |
| T <sub>b</sub> | LD    | +  | 36.1 ± 0.07 | 0.69 ± 0.17 | 0 ± 0.8    | 0.36 ± 0.17 | 0.2 ± 0.6  |
| T <sub>b</sub> | SD    | –  | 37.0 ± 0.12 | 0.85 ± 0.17 | 21 ± 0.9   | 0           |            |
| T <sub>b</sub> | SD    | +  | 36.6 ± 0.12 | 0.52 ± 0.16 | 17 ± 0.6   | 0.48 ± 0.17 | 4.0 ± 0.4  |
| V              | LD    | –  | 226 ± 18    | 72.7 ± 22.8 | 22 ± 1.9   | 92.7 ± 22.8 | 20 ± 0.7   |
| V              | LD    | +  | 218 ± 45    | 30.6 ± 64.5 | 5.0 ± 2.9  | 85.8 ± 64.6 | 17 ± 1.4   |
| V              | SD    | –  | 529 ± 64    | 160 ± 87    | 0.75 ± 0.4 | 143 ± 89    | 6.5 ± 1.1  |
| V              | SD    | +  | 403 ± 84    | 82 ± 11.7   | 11.4 ± 2.1 | 97 ± 11.2   | 1.7 ± 0.9  |
| 6-SMT          | LD    | –  | 0.93 ± 0.1  | 0.76 ± 0.2  | 3.1 ± 0.8  | 0.6 ± 0.2   | 3.0 ± 0.7  |
| 6-SMT          | LD    | +  | 1.3 ± 0.2   | 1.8 ± 0.3   | 5.0 ± 0.7  | 1.2 ± 0.3   | 5.3 ± 0.4  |
| 6-SMT          | SD    | –  | 4.4 ± 1.3   | 3.4 ± 1.2   | 1.4 ± 3.5  | 0.68 ± 1.2  | 20.3 ± 5.4 |
| SMT            | SD    | +  | 3.1 ± 1.8   | 2.4 ± 1.2   | 5.9 ± 1.8  | 0.34 ± 1.2  | 0 ± 1.0    |

<sup>1</sup>A. *russatus* mice were acclimated to either LD or SD in the absence (–) and presence (+) of A. *cahirinus* (AC). T<sub>b</sub> (in °C), V (in µL) and 6-SMT (in ng/hr) were measured at 3–4 hr intervals over 24–48 hr as described in the Materials and Methods section. Mean values were calculated for each time point and a baseline cosinor function was fitted to these mean values by nonlinear regression. The mean ± S.D. values of the variables obtained by the regression analyses are depicted. The fitted functions are presented as solid lines in Figs. 1–3. Mean ± S.D. values presented are for n = 8 in each group.

$$y(t) = M + A24 \cdot \cos [2\pi/24 (t - \phi 24)]$$

where t represents the time of day (in hr), M the mean of data points (mesor), A24 the amplitude of the rhythm,  $\phi 24$  the acrophase (the phase of the crest), and y(t) the predicted value of the variable at time t.

As most rhythms were sharply peaked or indicated more than one peak, the function fitted was the baseline cosine function (Ruf, '96):

$$y(t) = M + A24 \cdot \cos [2\pi/24 (t - \phi 24)] + A12 \cdot \cos [2\pi/24 (t - \phi 12)]$$

where t, M, A24, and  $\phi 24$  represent the respective values of time, mesor, amplitude, and acrophase of the circadian rhythm as above; A12 and  $\phi 12$  are the corresponding values of the 1st harmonics (12 hr), and y(t) the predicted value of the variable at time t. The results of the best fit regressions are represented as solid lines in the figures below and the values of parameters are given in Table 1.

### Statistical analyses

Data obtained from individual animals from each group were analyzed using either two-way

or one-way analysis of variance (ANOVA) followed by post-hoc Student's *t*-tests. Results of the statistical analyses are depicted in Table 2.

## RESULTS

Body temperature (T<sub>b</sub>) rhythms of A. *russatus* acclimated to long (LD) and short (SD) photoperiods are shown in Figure 1. In LD as well as SD animals, T<sub>b</sub> exhibited a daily rhythm, which could be fitted into a cosine function with a peak in the dark phase. In SD mice the mesor (M) of the T<sub>b</sub> rhythm increased by a mean of 0.5°C and the phase ( $\phi 24$ ) of the rhythm advanced by 3.3 hr compared to the respective values in LD mice, whereas the wave form and amplitude (A24) of the T<sub>b</sub> rhythm were unchanged. Exposure to A. *cahirinus* markedly modified T<sub>b</sub> waveforms in both SD and LD mice (Table 1). The T<sub>b</sub> rhythms were now more sharply peaked and fitted a baseline cosine function (Ruf, '96).

Two-way ANOVA performed on the individual body temperature data from the LD and SD animals indicated significant effects of exposure to A. *cahirinus* but not of the photoperiod. No interaction between photoperiod and exposure to A. *cahirinus* was found (Table 2).

TABLE 2. Statistical analyses (two-way ANOVA) of the data depicted in Figs. 1, 2, and 3 for significant effects of photoperiod and congener exposure with post-hoc *t*-tests on each factor

| Treatment variable | F                   | Photoperiod | Exposure | Photoperiod × exposure |
|--------------------|---------------------|-------------|----------|------------------------|
| T <sub>b</sub>     | F(3,238) = 2.99*    | NS          | *        | NS                     |
| V                  | F(3,158) = 13.53*** | ***         | *        | NS                     |
| 6-SMT              | F(3,164) = 17.66*** | ***         | NS       | NS                     |

\*\*\*P < 0.001; \*\*P < 0.01; \*P < 0.05; NS P > 0.05.

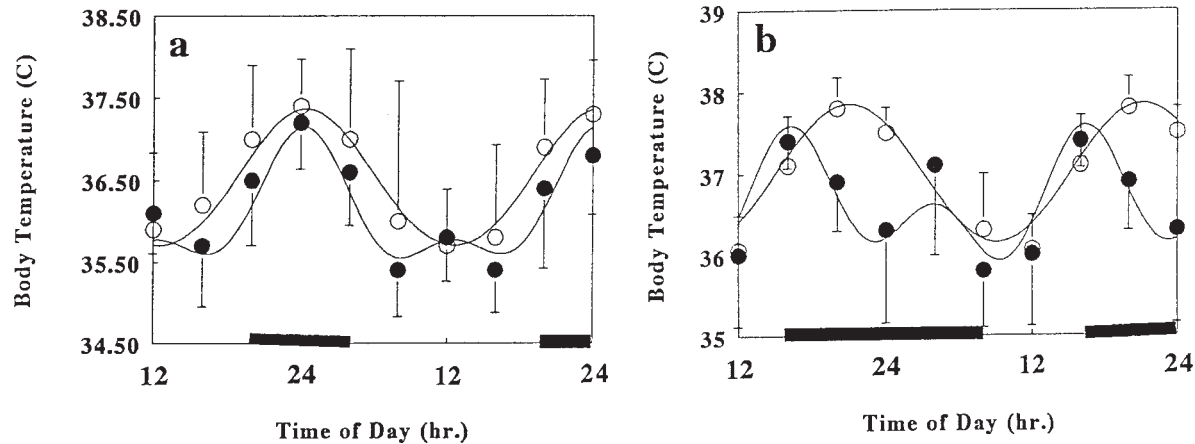


Fig. 1. Effects of *A. cahirinus* on body temperature ( $T_b$ ) rhythms in *A. russatus* acclimated to LD (a) or SD (b) before (○) and after (●) exposure to *A. cahirinus*. Mean  $\pm$  SD values of 8 animals are presented. The solid lines are theoretical curves

reconstructed from the best-fit parameters obtained by analysis of the data and presented in Table 1. Lights were on between 04:00 and 20:00 hr in LD, and between 08:00 and 16:00 hr in SD. The dark hours are depicted by a black line.

Urine volume rhythms (V) of LD and SD acclimated *A. russatus* are shown in Figure 2. In both LD and SD animals, there was a significant day-night rhythm in urine volume, with peaks at the beginning and end of the dark period, and minimal values during daytime. In the presence of *A. cahirinus*, the daytime V values increased under both photoperiods.

Statistical analyses indicated that the two variables (photoperiod and exposure) had significant effects on V rhythms. No interaction between photoperiod and exposure to *A. cahirinus* was found (Table 2).

The urinary excretion of the melatonin metabolite, 6-SMT, at various times of the day is shown

in Figure 3. A robust nocturnal secretion of 6-SMT was observed in the LD and SD animals. The duration of the 6-SMT peak was longer in SD than LD mice regardless of exposure to *A. cahirinus*.

Statistical analyses revealed that the 6-SMT rhythm was significantly affected by photoperiod and not by exposure to *A. cahirinus*. No interaction between photoperiod and exposure to *A. cahirinus* was found (Table 2).

The 6-SMT values for *A. russatus* at different hours of the day before and after exposure to *A. cahirinus* were compared (*t*-test). It was found that in LD mice, 6-SMT at 06:00 hr was significantly higher after exposure ( $P < 0.01$ ) whereas

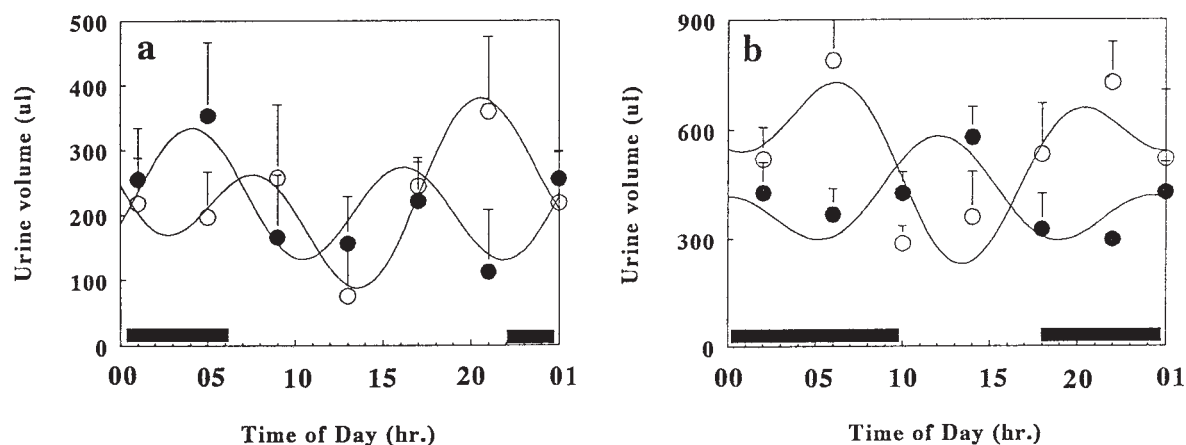


Fig. 2. Effects of *A. cahirinus* on urine excretion volume rhythms in *A. russatus* acclimated to LD (a) or SD (b) before (○) and after (●) exposure to *A. cahirinus*. Mean  $\pm$  SD values of 8 animals are presented. The solid lines are theoretical

curves reconstructed from the best-fit parameters obtained by analysis of the data and presented in Table 1. The dark hours are depicted by a black line.



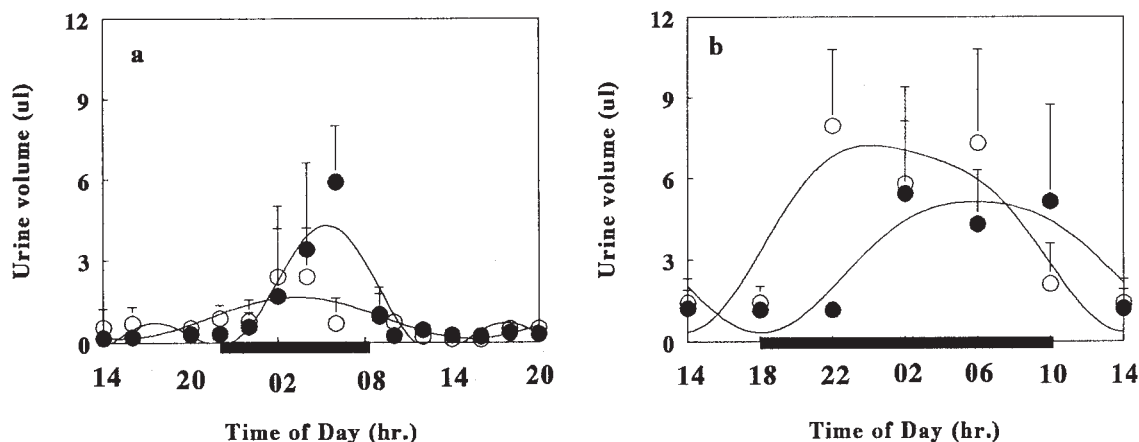


Fig. 3. Effects of *A. cahirinus* on 6-sulfatoxymelatonin rhythms in *A. russatus* acclimated to LD (a) or SD (b) before (○) and after (●) exposure to *A. cahirinus*. Mean  $\pm$  SD values of 8 animals are presented. The solid lines are theoretical

curves reconstructed from the best-fit parameters obtained by analysis of the data and presented in Table 1. The dark hours are depicted by a black line.

in SD mice, 6-SMT at 22:00 hr was significantly lower ( $P < 0.05$ ) compared to respective values before exposure.

$^3\text{H}$ -labeled 2DG accumulation by the SCN at various times of the day is depicted in Table 3. Representative autoradiographs of coronal brain sections from *A. russatus* that were injected with  $^3\text{H}$ -labeled 2DG are shown in Figure 4. 2DG uptake by the SCN of the LD, as well as of SD-acclimated *A. russatus*, was not significantly different at the early and late photophase. The uptake into the SCN was also not significantly different from uptake into adjacent areas at these hours. In LD mice exposed to *A. cahirinus*, 2DG uptake into the SCN tended to be enhanced at the late light phase (18:00 hr,  $P < 0.0001$ ,  $t$ -test). A similar tendency was found in SD acclimated mice at 14:00 hr.

TABLE 3. Uptake of 2-deoxyglucose (2-DG) by the SCN of LD and SD acclimated *A. russatus* in the absence and presence of *A. cahirinus* (AC)<sup>1</sup>

| Photoperiod | Time  | AC | Uptake          |
|-------------|-------|----|-----------------|
| LD          | 10:00 | –  | $0.16 \pm 0.1$  |
| LD          | 10:00 | +  | $0.22 \pm 0.02$ |
| LD          | 18:00 | –  | $0.27 \pm 0.01$ |
| LD          | 18:00 | +  | $0.49 \pm 0.03$ |
| SD          | 11:00 | –  | $0.25 \pm 0.21$ |
| SD          | 11:00 | +  | $0.11 \pm 0.1$  |
| SD          | 14:00 | –  | $0.14 \pm 0.1$  |
| SD          | 14:00 | +  | $0.30 \pm 0.18$ |

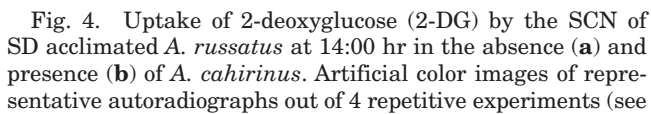
<sup>1</sup>Experiments were performed at the early and late photophase for LD and SD. Autoradiographs of the labeled slices were analyzed by image analysis and expressed in arbitrary units relative to the background values. Mean  $\pm$  S.D. values presented are for  $n = 4$  in each group.

## DISCUSSION

The results obtained here indicate that exposure to *A. cahirinus* induces significant changes in the  $T_b$  and V rhythms of *A. russatus* whereas melatonin production is not significantly affected.

The apparent changes in circadian rhythms in *A. russatus* may be an outcome of the increase in daytime activity that is provoked by the proximity to *A. cahirinus*. If so, the modified waveforms of the  $T_b$  and V rhythms may represent, at least in part, a direct stimulus-response (masking) effect of the enforcement of *A. russatus* to become more active during daytime, when its competitor is inactive. This hypothesis is compatible with the association of urine excretion with periods of activity demonstrated in rats (Cohn et al., '70; Windle et al., '92) and of elevated  $T_b$  with periods of activity demonstrated in *A. russatus* (Friedman et al., '97). It may also explain the tendency to increase uptake of 2-deoxyglucose (2DG) into the SCN in the late photophase.

A masking effect would imply that the modulation of circadian rhythms did not involve the underlying circadian pacemaker. However, this conclusion should be considered with caution because urinary excretion of the melatonin metabolite was measured at quite long intervals, and moderate changes in duration or phase of the rhythm might not have been detected. Moreover, in LD as well as SD mice 6-SMT in the end of the dark period was significantly higher after than before exposure to *A. cahirinus*, suggesting that a phase delay (or lengthening of the endogenous free-running period) of the endogenous clock has



occurred. Such a delay is compatible with the induction by forced activity of phase shifts of the daily activity-rest cycles in hamsters (Mrosovsky, '88; Van Reeth and Turek, '89).

Even if the clock has in fact been delayed by the exposure to *A. cahirinus*, not all rhythms have apparently been delayed. Some examples exist for the ability of conflicting time cues to internally dissociate the circadian system. In squirrel monkeys under artificial nonoverlapping light-dark (LD) and food availability (EF) cycles,  $T_b$  followed the LD whereas urinary potassium excretion followed the EF cycle (Sulzman et al., '78). In rats rendered diurnal by restricted-schedule feeding, the rhythm in vasointestinal peptide immunoreactivity in the SCN was advanced by 4 hr (Morin et al., '93). Internal dissociation of the cortisol rhythm has been observed in night shift workers (Weibel et al., '96). We have previously observed a significant correlation between  $T_b$  and activity rhythms in *A. russatus* that was modified by the photoperiod and by exposure to *A. cahirinus* (Freidman et al., '97). Thus, the apparent changes in the internal temporal order in *A. russatus* may reflect differences, besides masking effects, in coupling strength of individual physiological parameters with the underlying clock.

Interestingly, 2DG uptake by the *A. russatus* SCN during daytime was not different from that into the surrounding brain areas. This is unlike the situation in other species (Schwartz and Gainer, '77; Flood and Gibbs, '82), as well as in *A. cahirinus* (unpublished data) in which 2DG uptake into the SCN was more pronounced during the light period. There could be species differences or perhaps a low amplitude rhythm in the elec-

trical activity of the SCN in this animal, despite robust nocturnal behavior.

Despite the changes in daily schedules, the photoperiodic responses of *A. russatus* were maintained in the presence of *A. cahirinus*, as evidenced from the significant modification of 6-SMT and V in SD as compared to LD acclimation. In mammals, the photoperiodic message is conveyed by the duration of the melatonin peak (Reiter, '91). In the case of *A. russatus* the duration of the melatonin signal was maintained because the melatonin peak was not pushed into the daytime, at which time the synthesis would be acutely suppressed. We suggest that the nocturnal-type circadian system does not adapt to diurnality despite ages of forced diurnal activity. Consequently, temporal segregation, as a means for adaptive processes of animals to their habitat, is primarily a masking effect employed perhaps to maintain photoperiodic responsiveness which is crucial for seasonal adaptation.

## LITERATURE CITED

- Cohn CL, Webb D, Joseph D. 1970. Diurnal rhythms in urinary electrolyte excretion in the rat: influence of feeding habits. *Life Sci* 9:803-809.
- Flood DG, Gibbs FP. 1982. Species difference in circadian  $^{14}\text{C}$ -2-deoxyglucose uptake by suprachiasmatic nuclei. *Brain Res* 232:200-205.
- Friedman D, Haim A, Zisapel N. 1997. Temporal segregation in coexisting spiny mice (Genus *Acomys*): role of photoperiod and heterospecific odor. *Physiol Behav* 62:407-411.
- Haim A, Rozenfeld FM. 1993. Temporal segregation in coexisting *Acomys* species: the role of odor. *Physiol Behav* 54:1159-1161.
- Harrison DL, Bates PJJ. 1991. The mammals of Arabia. Kent Harrison Zoological Museum.
- Moore-Ede MC, Sulzman FM. 1981. Internal temporal order. In: Aschoff J, editor. *Handbook of behavioral neurobiology*, volume 4. New York: Plenum Press. p 215-241.

- Morin AJ, Denoroy L, Jouvét M. 1993. Daily variations in concentration of vasoactive intestinal peptide immunoreactivity in hypothalamic nuclei of rats rendered diurnal by restricted-schedule feeding. *Neurosci Lett* 152:121–124.
- Mrosovsky N. 1988. Phase response curves for social entrainment. *J Comp Physiol A* 162:35–46.
- Reiter RJ. 1991. Pineal melatonin: cell biology of its synthesis and of its physiological interactions. *Endocr Rev* 12:151–180.
- Ruf T. 1996. The baseline cosine function: a periodic regression model for biological rhythms. *Biol Rhythms Res* 27:153–165.
- Schwartz WJ, Gainer H. 1977. Suprachiasmatic nucleus: use of  $^{14}\text{C}$ -labeled deoxyglucose uptake as a functional marker. *Science* 197:1089–1091.
- Shkolnik A. 1971. Diurnal activity in a small desert rodent. *Int J Biometeorol* 15:115–120.
- Sulzman FM, Fuller CA, Moore-Ede MC. 1978. Comparison of synchronization of primate circadian rhythms by light and food. *Am J Physiol* 234:R130–R135.
- Van Reeth O, Turek FW. 1989. Stimulated activity mediates phase shifts in the hamster circadian clock induced by dark pulses or benzodiazepines. *Nature* 339:49–51.
- Weibel I, Spiegel K, Follenius M, Ehrhart J, Brandenberger GA. 1996. Internal dissociation of the circadian markers of the cortisol rhythm in night workers. *Am J Physiol* 270:E608–E613.
- Windle RJ, Forsling ML, Guzek JW. 1992. Daily rhythms in the hormone content of the neurohypophyseal system and release of oxytocin and vasopressin in the male rat: effect of constant light. *J Endocrinol* 133:283–290.